Dynamic programming (class 6)

1. Is the following recursive algorithm effiecient for calculating the nth Fibonacci number? If not, give better.

   FIB(n)

   if $n \leq 1$ return n

   else return FIB($n-1$)+FIB($n-2$)

2. (a) Give a $O(n^3)$ DP algorithm for finding the optimal paranthesization of n matrices of sizes $r_{i-1} \times r_i$ $(1 \leq i \leq n)$.

   (b) Run the algorithm for $n = 6$ and sizes 25x30, 30x10, 10x5, 5x10, 10x15, 15x20.

3. (a) Give a $O(nm)$ DP algorithm for finding one of the longest common subsequences of $X = (x_1, x_2, \ldots, x_n)$ and $Y = (y_1, y_2, \ldots, y_m)$

   (b) Run the algorithm for $X = (a, b, b, c, d)$ and $Y = (c, a, e, d, b, a)$.

   (c) Run the algorithm for $X = (0, 1, 1, 0)$ and $Y = (1, 1, 0, 1, 0, 1)$.

4. Give a $O(n^2)$ an algorithm for finding the longest monoton increasing subsequence of an n element sequence.